Yilin Wang

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My Homepage

Education

- **University of Alberta** — Edmonton, Canada
  PhD of Electrical and Computer Engineering Department; GPA: 3.67/4.0 — *Sep. 2023 - Now (Transferred)*
  Courses: *Techniques of Maths for Stats (*course for STAT PhD student)*
  Supervisor: *Li Cheng*

- **University of Alberta** — Edmonton, Canada
  M.S.c of Electrical and Computer Engineering Department; GPA: 3.9/4.0 — *Sep. 2021 - Aug.2023*
  Courses: *Fuzzy Sets Human-Centric System, Computer and Robotic Vision, Deep Learning in Computer Vision, Statistical Learning, Theoretical Foundations for Reinforcement Learning*
  Supervisor: *Li Cheng*

- **Beijing Institute of Technology** — Beijing, China
  B.S. of Electrical Information Engineering; GPA: 3.8/4.0 — *Aug. 2017 - Jul. 2021*

Employment

Associate Researcher, Intern — Jun. 2026 - Now
Reality Labs, Meta Inc. — Mentor: Petr Kadleček

- 3D Human Motion Generation; Motion Personalization; Body Tracking.

Associate Researcher, Intern — Oct. 2023 - Jun. 2024
Noah's Ark Laboratory, Huawei Technologies Canada Co. Ltd — Mentor: Juwei Lu

- Motion Synthesis; Character Animation; Single-view 3D Object Reconstruction; Gaussian Splatting.

Associate Researcher, Intern — Apr. 2023 - Sep. 2023
Noah's Ark Laboratory, Huawei Technologies Canada Co. Ltd — Mentor: Jun Jin, Xi Chen

- LLM Finetune; VLM; Robot Foundation Models.

Publications

- **Yilin Wang**, Chuan Guo, Yuxuan Mu, Muhammad Gohar Javed, Xinxin Zuo, Li Cheng, Hai Jiang, Juwei Lu. *"MotionDreamer: One-to-Many Motion Synthesis with Localized Generative Masked Transformer."* In International Conference on Learning Representations (ICLR). 2025. *(Accept rate: 26.65%)*

- **Yilin Wang**, Chuan Guo, Li Cheng, Hai Jiang. *"RegionGrasp: A Novel Task for Contact Region Controllable Hand Grasp Generation."* European Conference on Computer Vision Workshops (ECCVW). 2024.

- Siyuan Li, Weiying Chen, **Yilin Wang**, Xinxin Zuo, Xingyu Li, Li Cheng. *"Progressive Pose-Guided 4D Animal Reconstruction from Monocular Video."* European Conference on Computer Vision (ECCV). 2026. *(Spotlight -Accept rate: 1.60%)*

- Yuxuan Mu, Xinxin Zuo, Chuan Guo, **Yilin Wang**, Juwei Lu, Xiaofeng Wu, Songcen Xu, Peng Dai, Youliang Yan, Li Cheng. *"GSD: View-Guided Gaussian Splatting Diffusion for 3D Reconstruction."* European Conference on Computer Vision (ECCV). 2024. *(Accept rate: 27.9%)*

- Jianyuan Yin, Zheyi Fan, Shuai Chen, **Yilin Wang**, *"In-Depth Exploration of Attribute Information for Person Re-Identification"*, APPLIED INTELLIGENCE. 2021.

Highlight Projects

Group Dance Choreography Synthesis (Under submission)
University of Alberta — Jan. 2025 - March. 2026

- Aimed to generate high quality dance choreography for groups of multiple dancers in interaction driven by music.
- Establishing an expressive masked auto-regressive model for learning the underlying dance motion patterns and interaction cues between multiple dancers in a group.
- Working on extended downstream applications with optional control signals, such as group dance formation graph, leader dance motions, and dance style labels.

One-to-Many Motion Synthesis
University of Alberta & Huawei Technologies Canada Co. Ltd — Nov. 2023 - Sep. 2024

- The topic is to generate natural and diverse motions given a single reference motion with arbitrary skeletons, while preserving the reference local motion patterns.

- A localized vector quantization based mask transformer is proved to be a strong paradigm in this specific topic where the training data includes only one motion sequence.
- Downstream applications demonstrated, including crowd animation, editing, and beat-aligned dance synthesis.

Region Controllable Hand Grasp Generation
University of Alberta Aug. 2022 - Mar. 2023

- A new task of region-based controllable hand grasp generation is proposed, in which generated hand grasps are expected to contact with a selected surface region of a object by the thumb finger tip.
- A conditional-VAE based generative model with a region-aware training strategy, and an interaction-aware transformer module.

Honors and Awards

- Outstanding Graduate of Beijing Institute of Technology July. 2021
- The Second prize, 2019 Beijing Integrated Circuit Design Competition for College Students Nov. 2019
- Champion(top 1%), the Beijing Institude of Technology Integrated Circuite Design Competition Jun. 2019
- The Second prize, Beijing Mathematical Contest in Modeling for College Students Jun. 2019
- First-level Scholarship Renmin Scholarship (x 1 times) Jan. 2018
- Second-level Scholarship Renmin Scholarship (x 5 times) 2018 - 2021